

MOCK amplifier design

EE210

Design an amplifier with the following specifications.

Load resistance, $R_L = 1\text{ k}\Omega$. One of the terminals of R_L must be grounded, and the other terminal must be AC-coupled with the rest of the network.

$HD_1 / HD_3 = 40\text{ dB}$ for output pk-pk swing of 1V when a sinusoidal input of frequency 100 MHz is applied. Use of an inductor is not allowed. The maximum size of a capacitor is 100pF. There is no specification on the input swing, and the number and type of transistors that you can use.

A single supply $V_{DD} = 1.8\text{V}$ is available for you, and the input sinusoidal signal is centred around zero DC.

Note: The specification does not say anything about the strength of the input tone.

HD_1 refers to the strength of the fundamental tone, and HD_3 refers to the strength of the third harmonic. (You can look into the spectrum of the signal to find this out)

The first thing that you should do is to learn the software. We will be using Cadence, the setup details and the start-up guide is available in the attachment. You need a windows environment setup in which the virtual box (pre-loaded with cadence and the technology node) can run.

The first thing to do is to characterize the transistor (i.e. find the V_{th} , I-V char or the g_m / I_D char). The following video shows you a way to go about doing it. You will notice that while designing, we don't generally try to find the mobility or the C_{ox} explicitly. We work at a higher abstraction and try to characterize the g_m directly (after all that's the most important parameter).

<https://www.youtube.com/watch?v=DlxgkM2zfNU>

Submit your schematic, the test-bench and the test-setup which we can run at our end.